In terms of the scientific method, how does astronomy differ from a lab science like chemistry or biology?

The main difference is in manipulation of variables. A 'lab science', like chemistry or biology, centers on control experimentation. It is possible for the scientist to mold a substance, increase the temperature of a test tube, or remove an isolated protein and see what happens. The scientist can set up a "control group" and confirm their results under a specified set of parameters. Astronomy, however, is almost entirely an observation science. A scientist cannot alter a star, nor can they create a "control" galaxy to determine what happens if its dark matter is removed. As they do not create conditions from scratch, rather, they "use the universe as their laboratory". Astronomers wait for nature to present them with evidence to analyze, in the form of observable phenomena that have been occurring throughout space and time.

How do astronomers know what objects at far distance from earth are like?

We have the following ways of confirming that we understand far objects:

1. Data-high-quality, empirically gathered facts, such as those transmitted by Space Telescopes such as Cassini or the Large Ground-Based Telescopes, need to be accounted for.
2. Logic-if physics works for objects in our local solar system, we can logically infer it will work for objects further out; e.g., if physics dictates that a planet will orbit a star as it does Earth around the Sun, then we infer it is applying for all other orbiting planets too.
3. Reproducibility-once two or more independent scientists reach the same conclusion then the understanding, they arrive at can be trusted further. We understand remote objects such as these by confirming models against an enormous quantity of data that is seen to match all aspects of the theoretical prediction.

Regarding the argument that structures like Stonehenge or the Great Pyramids are evidence of "ancient astronauts," I would argue that this perspective underestimates human history and ignores the documented astronomical prowess of our ancestors.

The answer *'Human Ingenuity'*

The thought that humans required alien assistance to build these structures is the result of a general disdain for ancient intellect and geometry. The Ancient Greeks for example managed to calculate that the Earth was a sphere, and estimate its circumference with impressive accuracy using just shadows and a basic understanding of geometry. If they were capable of working out the size of a planet without ever leaving the ground, then it is not a huge leap to suppose that they or another civilization could point stones or bricks to the stars using similar precise observations.

The Sky Was a Workhorse Tool

The ancient world relied on the sky in a number of ways other than for sheer wonder. The sky was a calendar, clock and compass. Survival in an agricultural world relied heavily on having an accurate calendar of solar, lunar and stellar movements so that crops could be planted or herds driven to fresh pasture, and there is no doubt that Stonehenge was one very sophisticated calendar. These sites are not alien works but the physical representation of many thousands of years of observation.

Occam's Razor and The Evidence.

There is zero evidence in these sites, either directly or indirectly, to support anything like alien intervention; there is not one alien tool, one trace of alien metals, or one atom of non-terrestrial waste found there. There is however evidence of quarries, of copper tools and of immense human manpower put to use. To argue alien interference is simply to make inspired guesswork with no supporting facts whereas that the buildings are evidence of human endeavor is an argument backed up by both archaeological evidence and cultural needs.

By the claim that these monuments were the work of extraterrestrial intervention we deny our ancestors their place in history. The same '*patterns in nature*' that we study now were then worked into the very fabric of civilization.